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Quantum computing empowering blockchain technology with post quantum resistant cryptography for multimedia data privacy preservation in cloud-enabled public auditing platforms

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Abstract
The multimedia environment has undergone significant growth, particularly in the area of multimedia data and its migration to cloud platforms, which has raised issues about security, confidentiality, data integrity, and privacy protection. While Blockchain Distributed Ledger Technology (BDLT) offers decentralized trust and transparency the advent of Quantum Computing threatens classical cryptographic primitives, which make multimedia data increasingly vulnerable. This paper proposes a novel and secure framework that collaborates BDLT with quantum-resilient, mainly known post-quantum cryptographic schemes to ensure long-term data integrity and privacy preservation in cloud-based infrastructures. Due to this, the proposed solution enables secure, efficient, and transparent that helps in public auditing of multimedia content without compromising stakeholder confidentiality. It leverages Zero-Knowledge Proofs (ZKPs), lattice-based cryptography, and smart contract automation, which model fortifies data authenticity verification against quantum attacks. Simulation results illustrate the effectiveness of the proposed framework that achieves a 98.21% accuracy in data integrity verification, a 96.84% reduction in quantum vulnerability, and an 87.85% efficiency gain in auditing speed compared to classical BDLT-enabled platforms. In addition, privacy leakage in multimedia systems is reduced by 92.47% proving the framework's robustness. This solution underscores the potential of synergizing BDLT, quantum secure cryptography, and cloud computing to build a future-proof solution for privacy-protected multimedia data management and public auditing.

Keywords Multimedia systems, Blockchain, Cloud computing, Quantum computing, Post quantum resistant cryptography, Privacy preservation

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Introduction

The increasing use of multimedia content, especially test data, images, and videos as rich data streams, has led to a greater reliance on cloud computing systems, mainly for real-time access, scalable storage, and worldwide sharing [1, 2]. However, cloud infrastructure offers remarkable flexibility and resource efficiency, which raises serious concerns about data security, privacy, and integrity, especially in situations where third parties are involved in ensuring that multimedia contents are preserved for public access [1, 3, 4]. Although they are somewhat successful, classical cryptographic algorithms—which are primarily used to safeguard data—are gradually losing their effectiveness in the face of new computer paradigms, chief among them being Quantum Computing (QC).

With their unmatched capacity to execute intricate calculations that are exponentially more reliable than those of classical computing systems, QC directly threatens traditional encryption techniques like Rivest-Shamir-Adleman, Elliptical Curve Cryptography, and Digital Signature Algorithms [5, 6]. The majority of current digital security methods are built on these algorithms, and cloud-based multimedia data is at previously unheard-of risk due to their susceptibility to quantum attacks. In this manner, post-quantum cryptography, especially the lattice-based schemes, has emerged as a promising direction to achieve quantum resistance [7–9], as shown in Fig. 1. Nevertheless, integrating post-quantum cryptography into cloud architecture alone is not sufficient to ensure transparency and trust in the data auditing mechanism.

However, Blockchain Distributed Ledger Technology (BDLT) offers a decentralized, unforged, and tamper-proof ledger capable of enhancing or improving transparency, traceability, and immutability in data organization and management systems [9–11]. In order to enable smart contracts and audit trails into BDLT-based Distributed Applications (DApps), organizations can automate compliance, reinforce confidentiality, and integrity, and mitigate unauthorized access. Along with that, when harmonized with post-quantum cryptography and public auditing protocols, BDLT can serve as a secure intermediary that not only maintains auditability but also preserves user security and privacy through cryptographic Zero-Knowledge Proofs (ZKPs) with privacy-preserving access control mechanism [12–14].

As a result of this assessment, this paper offers a novel, scalable, and secure framework that integrates post-quantum cryptography, BDLT, and quantum-aware security protocols to address the issues of privacy, confidentiality, integrity, and transparency in a public auditing environment that is enabled by cloud computing, particularly for multimedia data management, as shown in Fig. 2. In order to create a cryptographic scheme, the suggested solution uses the fundamental strategies of lattice-based cryptography, which are mathematical problems involving lattices, such as essentially grids of points in high dimension space, enhanced by BDLT immutability and smart contract automation. It resists breaches based on quantum technology and supports transparent audits.

On the other end, experimental validations demonstrate that the system achieves superior performance across different dimensions, such as 98.31% accuracy in

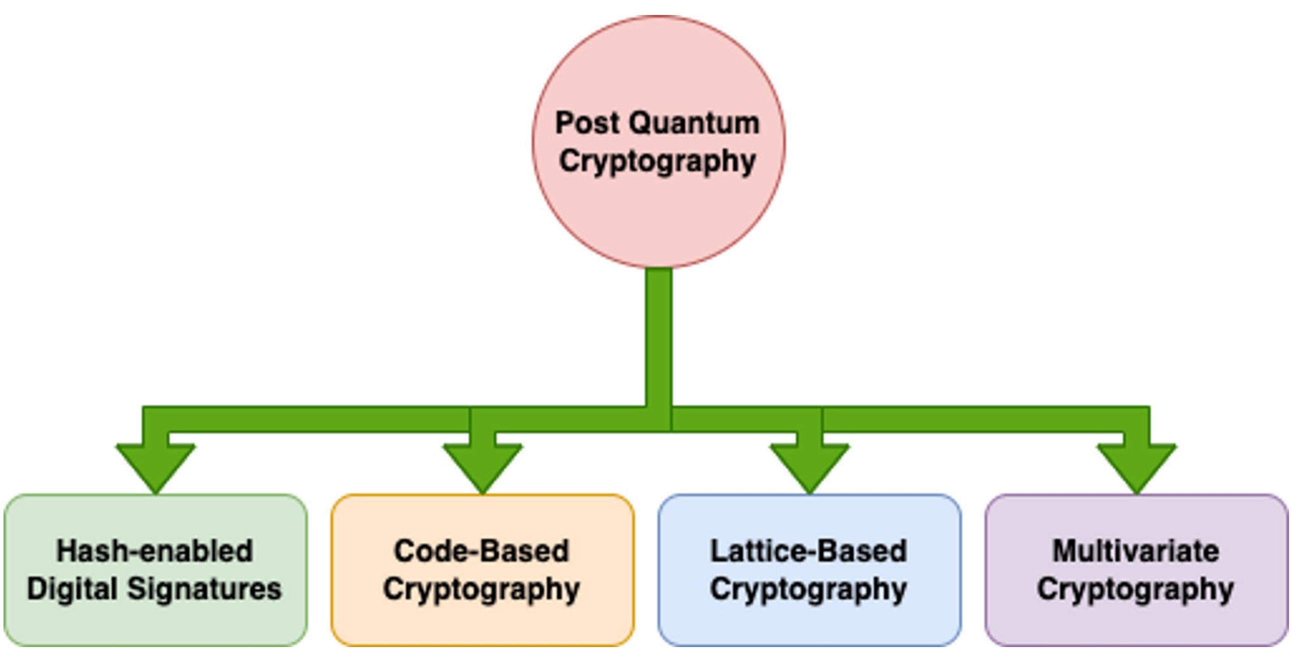

Fig. 1 Post-Quantum Cryptography's Working Objectives

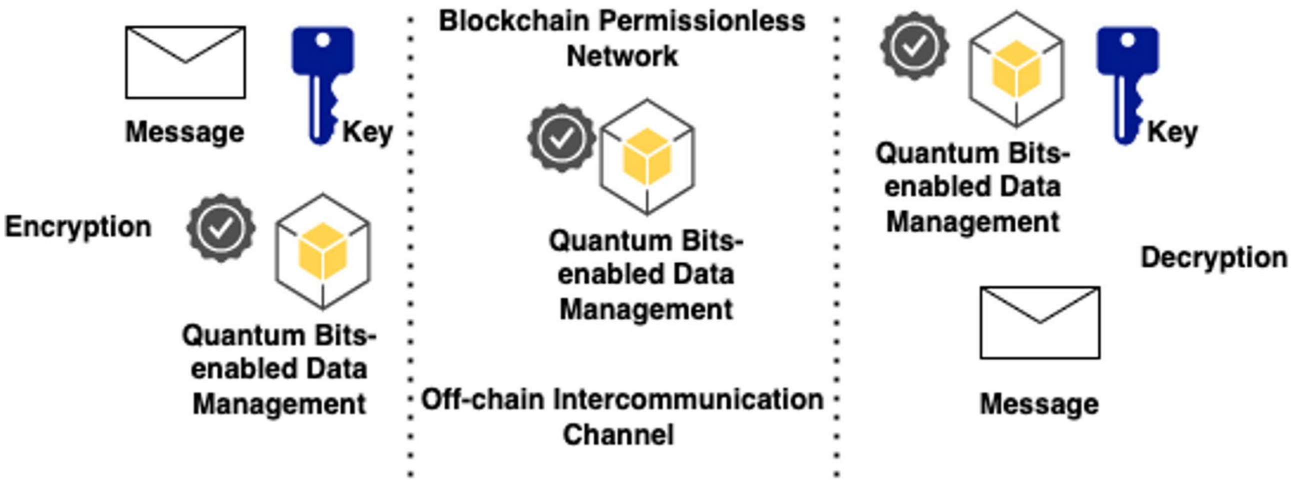

Fig. 2 BDLT-enabled Data Protection in Public Network

data confidentiality and integrity verification, 96.84% decrement in vulnerability to quantum attacks, and 87.65 enhancement in auditing throughput. More than that, 92.47% decrement in multimedia data leakage. Such results underscore the potential of a futuristic framework that fulfills the evolving demands of protected data management in an era poised for quantum dominance.

Motivation and Objective of this Research.

The main source of motivation is the assessment of multimedia data, which we see in the digital age as including images, audio, videos, hypertext, and linked streaming, or hypermedia, which makes up a significant amount of global data preservation and processing in the cloud environment. Industrial-based developments, especially from healthcare and digital forensics to smart monitoring and e-governance, increasingly rely on cloud computing for ubiquitous access and storage-based scalability. Throughout this, the major convenience comes with rising concerns about data privacy protection, confidentiality, integrity, transparency, provenance, and unauthorized access controls, particularly during the third-party public auditing process. However, the list of public auditing is highlighted as follows:

- Verifying the circulation figures of publications.
- Monitoring government spending and operations.
- Promote responsible advertising practices.

In contrast, BDLT has become a promising integration option because of its unforged environment, decentralized trust, and tamper-resistant authentication and data verification. Its current reliance on the hash-encryption technique based on classical cryptography, however, has recently exposed it to a serious and difficult problem. Among other things, it is susceptible to attacks using quantum computing. It is worth noting that the QC rapidly advancing the cryptographic protocols, such as the updating occurrent in the hierarchy of Rivest-Shamir-Adleman, Elliptical Curve Cryptography, and Digital Signature Algorithms, along with Hash Encryption (SHA) that are expected to become more obsolete, due to posing imminent threat to the integrity and confidentiality of cloud-enabling storage and multimedia data. In addition, multimedia data is inherently more sensitive and voluminous than textual data, which makes the current system more susceptible to privacy breaches, just like deepfake manipulation, and distribution of unauthorized access. By evaluating all such prospects, the lack of quantum-resilient auditing mechanisms in the current BDLT-enabled cloud model leaves a critical research gap in securing multimedia data governance.

Nonetheless, the need to create a quantum safe audit trail system that is future-proof and protects privacy is what drives this research. To create a privacy-protected cryptographic environment, BDLT transparency and post-quantum cryptographic resilience must be combined to create a scalable and safe applicational hierarchy that upholds public data auditing's credibility, as show in Fig. 2. ZKPs aid in lightening the authentication process throughout this entire process without disclosing private multimedia content. As a result, filling in these gaps is not just a quick fix but also necessary to guarantee that contemporary digital infrastructures are resilient in the face of the impending QC age.

Research Questions and Contributions.

The following is a list of the research questions covered in this paper:

1. How can BDLT effectively integrate with post-quantum cryptography algorithms to secure multimedia data preservation in a cloud-enabling environment against quantum threats?

2. How can a BDLT-enabled public auditing framework be designed to ensure data integrity and transparency?
3. How effective does the proposed integration provide reliability in terms of mitigating potential vulnerabilities in a quantum-aware threat environment?
4. How smart contract-enabled access control functions ensure transparency, dynamic user permission, and compliance in a secure auditing environment, especially public auditing.

The major contribution of this paper is discussed as follows:

- In order to provide long-term defence against QC attacks, this research integrates lattice-based post-quantum cryptographic algorithms into a revolutionary cloud-enabled multimedia data integrity, confidentiality, and privacy framework.
- This paper provides an effective and efficient decentralized BDLT-enabled public auditing environment using smart contracts with a Markle hash tree structure for making a tamper-proof audit trail and creating transparency for multimedia content.
- This proposed Distributed Application (DApp) incorporates ZKPs and homomorphic encryption to allow third-party auditors to verify data integrity without accessing the actual content.
- A quantum-aware threat modelling is calculated and created within the DApp that accounts for both classical and quantum attack vectors for providing a robust security posture.
- Due to this, the proposed framework achieves significant improvements with 98.31% accuracy in data confidentiality and integrity verification, 87.65% gain in audit efficacy, and 92.47% reduction in data leakage, validated through extensive simulation of multimedia data.
- This paper implements DApp with fine-grained access control policies via smart contracts enabling dynamic permissions for users and auditors, which promoting data governance in real-time.

A. *Outline of the Study.*

This paper's reminder is organised as follows: A thorough description of the relevant research and underlying technologies based on BDLT and Quantum Computing (QC) is provided in Sect. Related work. On the other hand, the suggested structure, technique, and working sequences for carrying out activities are presented in Sect. Method and material. The experimental setup and implementation are covered in Sect. Simulation, results, discussion, along with the results and their assessments. Section Implementation challenges and limitations highlights, marks, and discusses the list of implementation problems and challenges along with some potential fixes. Lastly, the conclusion in Sect. Conclusion and future direction brings this paper to a close.

Related work

In order to address contemporary issues with data security, privacy protection, and secure preservation, the considerable confluence of cloud computing, BDLT, and post-quantum encryption has gained traction [13–15]. In order to provide an open public auditing environment, the aforementioned sections examine the most recent research in the areas of cloud-enabling BDLT technology, maintaining hierarchies, and multimedia privacy security.

A. *Cloud Computing and Multimedia Data Protection.*

It is worth noting that cloud computing provides an on-demand, scalable, and pay-per-use model for storing, processing, and infrastructural facilities for evaluating multimedia content [15, 16]. However, the transmission of multimedia data to third-party cloud environments raises critical privacy and integrity concerns. The papers [17] and [18] mentioned vulnerabilities, including data forgery, tampering, information leakage, manipulation, and unauthorized access-related raising problems. There are various approaches involved in access control, encryption, and watermarking that have been proposed but often fall short in terms of scalability and computational efficiency. It is also due to misinterpretation when dealing with high-resolution media files.

On the other side, BDLT-enabled unforged data management properties have been recognized as one of the crucial solutions for ensuring trust and accountability in cloud data storage. The paper [19] introduced BDLT-enabled auditing mechanisms using Merkle trees and smart contracts to validate data confidentiality and integrity without sharing the content, even in a partial manner. In a similar manner, the paper [20] proposed a BDLT-enabled DApp for a secure sharing and exchange environment for digital evidence in the process of forensics investigation. In addition, such approaches generally depend on classical cryptographic primitives that leave them susceptible to the current quantum threats.

B. Post-Quantum Cryptography.

Post-quantum cryptography requires implicating cryptographic algorithms that are secure against quantum attacks. In this whole scenario, the Lattice-based model, like N-th degree Truncated polynomial Ring Units, has been widely utilized as previous research presented. It is because of the provable hardness assumption and creating efficiency in key generation and encapsulation, along with digital signature [21, 22]. The working hierarchy of post-quantum cryptography presented by IBM is becoming the standardized process that has endorsed such mechanism as a strong candidate for making reliable quantum-resilient cryptography [22, 23]. While research in post-quantum cryptography is becoming more and more extensive due to the strong integration with BDLT and cloud-enabling platforms for multimedia content; till now, it is in a nascent stage.

Nonetheless, third-party participation is permitted or mandated by the present public auditing systems to confirm and examine the integrity of cloud data without having access to its contents. One of the articles covered the well-known new methods for ensuring privacy, security, and dependability, like homomorphic encryption, provable data possession, and ZKPs [24]. It is important to note that the paper [25, 26] presented auditing systems that protect the confidentiality of multimedia materials. On the other hand, the scalability debate is brought to light [27–29] because it is still a difficult problem, especially when it comes to large-scale multimedia material.

However, a few recent studies [30–32] have suggested cooperative methods by combining BDLT with cryptographic techniques such as NuCypher Threshold Re-Encryption Mechanism and Hash Encryption (SHA-256 and SHA-512), which are thought to be very few marked or addressed as the specific need of multimedia data in quantum-aware content. Many Dapps that employ the aforementioned approach may not have strong privacy preservation safeguards during auding, or they may ignore the consequences of quantum assaults. Different research gaps are taken into consideration while creating a comprehensive architecture that incorporates BDLT, post-quantum cryptography, privacy-protected audit protocols, and smart contracts designed for cloud environments with multimedia [32, 33].

Method and material

Testing the proposed solution on many datasets while adhering to the general criteria of transactional scheduling, delivery, block management, post-quantum cryptographic computation, ledger management, ledger optimisation, scaling, throughput fluctuation, and latency is how this work's performance is evaluated. The datasets list includes the following:

There are different datasets are used in order to evaluate the proposed solution's reliability and effectiveness in every set of public auditing, in this paper we make a successful test for the proposed simulations to calculate third-party accessing in terms of an unforged, tamper proof, and non-manipulated in multimedia content while the process of auditing is running. However, the list of datasets and related descriptions is mentioned as follows:

1. **Dataset #1**: Multimedia Signal Processing and Understanding Lab's Dataset (Link: https://mmlab.disi.unitn.it/home).
   a. More than a thousand multimedia data points collected from more than different connected devices that make up this Multimedia Lab Dataset of MMLAB.
2. **Dataset #2**: SWS 2013 Multimedia-enabled Dataset Based on Query by Example Keywords (Link: https://speech.fit.vut.cz/software/sws-2013-multilingual-database-query-by-example-keyword-spotting).
   a. This multimedia data consists of over 10,000 multimedia data points gathered from over a thousand connected devices.
3. **Dataset #3**: YLI-GEO Placing Task Dataset proposed by multimedia commons (Link: https://multimediacommons.wordpress.com/yli-geo-placing-task-datasets/).
   a. More than 15,000 multimedia data points were collected from more than 1,000 linked devices to create this multimedia data.
4. Dataset #4: LIRIS-ACCEDE Dataset proposed by ECOLE CENTRALE LYON (Link: https://liris-accede.ec-lyon.fr).
   a. This multimedia data was created by gathering over 4,000 multimedia data points from over 1,000 connected devices.

A. *Problem Formulation and Description.*

It is evident that there is a distinct research gap in the implication of a holistic framework that integrates BDLT-enabled smart contracts, cloud-enabling privacy preservation, and post-quantum cryptography for making audit protocols tailored for multimedia systems. However, this section highlights the working steps of the proposed framework through mathematical notation and cryptographic constructs. It is due to the strong presence of multimedia data preserved in cloud infrastructure, verifiable through a BDLT-enabled post-quantum secure auditing protocol.

The multimedia data 'Md' is the first encrypted using symmetric key cryptography 'C' (as mentioned and

Table 1 Notations

Notation	Explanation
Md	Multimedia Data
C	Cryptography
Encryp	Encryption
Ck	Change in Cryptographic
H	Hash Encryption
T	Time
BDLT	Blockchain distributed ledger technology
sc	Smart contract
PM	Proof Management
ZKPs	Zero Knowledge Proofs

explained in Table 1). The designed equation is designed as follows:

C = Encryp [symmetric key (Md)] Eq. (1)

This symmetric key is then protected using a lattice-based post-quantum encryption mechanism, like Dilithium illustrated as follows:

*Ck = Post Quantum Encryp * Post Quantum Key Generation (symmetric key) Eq. (2)*

The integration of 'C' and 'Ck' are preserved in the cloud, while their hash digest is recorded on the BDLT in accordance with the following criteria (as also evaluated in Fig. 3):

hash (Md) = H (C | Ck | T) Eq. (3)

However, a BDLT-enabled smart contract logs the transaction with as follows:

BDLT = {hash (Md), cloud addresses, T} Eq. (4)

Whereas the smart contract 'sc' is threshold during public auditing and validation in terms of time consistency and data integrity, as shown in Fig. 3.

*H {C * Ck * T} = hash (Md) Eq. (5)*

On the other end, utilizing a lattice-based digital signature algorithm, such as Dilithium, where the data owner signs the hash data as follows:

Sigma = Signature (secure key (hash(Md))) Eq. (6)

In the process of verification and validation:

Verification = Post-Quantum Key Generation (hash (Md, Sigma)) Eq. (7)

This ensures confidentiality and integrity even under quantum adversarial models. In order to allow a third-party auditor to evaluate and validate the data without accessing 'Md', a non-interactive ZKPs is generated, which is considered a promising result, along with provide a lightweight and cost-efficient environment as follows (as shown in Fig. 4):

Proof Management (PM) = ZKPs (hash (Md), Sigma) Eq. (8)

Through evaluated and validated the public audit (as illustrated in Fig. 4):

*ZKPs * Verification/Validation (hash (Md), PM) Eq. (9)*

This tuning and their adaptation guarantee confidentiality and integrity while still ensuring accountability. The major constraints that need to be examines is mentioned as follows:

1. A (Integrity) = Accuracy of integrity verification.
2. R (Quantum) = Reduction in quantum vulnerability.

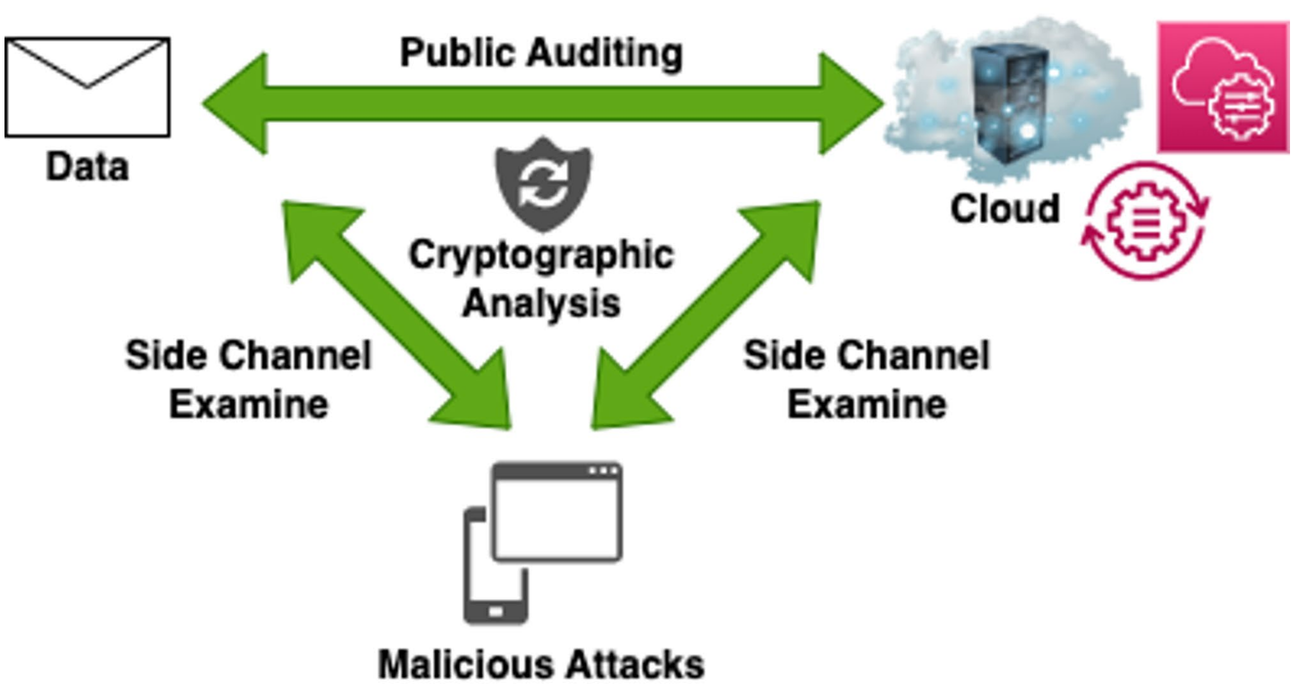

Fig. 3 Procedure of Cryptographic Analysis

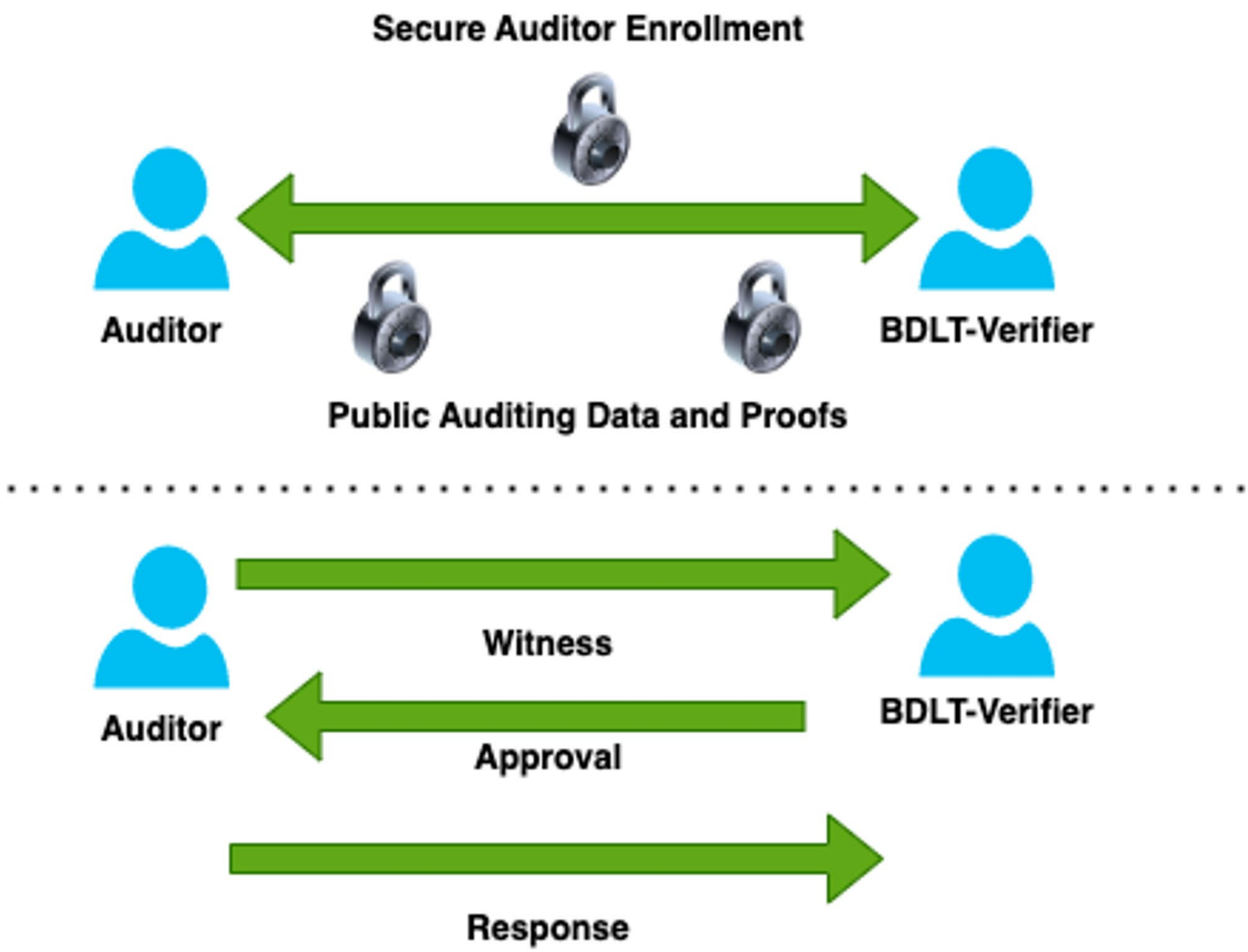

Fig. 4 Operational Execution of ZKPs

3. E (Audit) = Efficiency gain in audit trail and throughput.
4. L (Privacy) = Leakage reduction in multimedia content and privacy.

Then, we receive the simulation results as follows:

A(Integrity) = 98.31%, R (Quantum) = 96.84%, E (Audit) = 87.65%, and L (Privacy) = 92.47%.

B. Proposed Framework

First, Fig. 5 shows the phases involved in the proposed framework's operation, emphasising the function of BDLT. The suggested node-enabled block-based transactions are checked and confirmed to offer a safe and secure setting for public auditing, particularly for multimedia data scenarios. The auditor is able to participate in the review process through this approach. The transaction processor is essential for controlling probabilities and establishing communication between block data, as shown in Fig. 5. But while the blocks are being rescheduled or new transactions are being added, the REST API is always running. On the other side, block verification necessitates the creation of two intercommunication channels—on-chain and off-chain—in order to manage implicit and explicit transactions. In contrast, ZKPs set up access controls between BDLT specialists and auditors for the purpose of public auditing. Compared to other cutting-edge authentication or access control systems, it is a dependable, affordable, and lightweight solution. Hash-based digital signatures are essential to controlling ledger security and protection during this procedure. Every stakeholder involved is joining and carrying out transaction executions in accordance with the proposed DApp's consensus rule. Due to the possibility of consensus, a majority of 51% of participants must vote in favour of any changes to the prior transaction or update.

On the other hand, as illustrated in Fig. 6, the general layout and operation of the suggested framework are separated into four main areas: data encryption and access control; evaluation of sensitive data from encryption to decryption; public auditing of data and proof using ZKPs; and BDLT integration of ledger protection and cloud preservation management. The public auditing platform

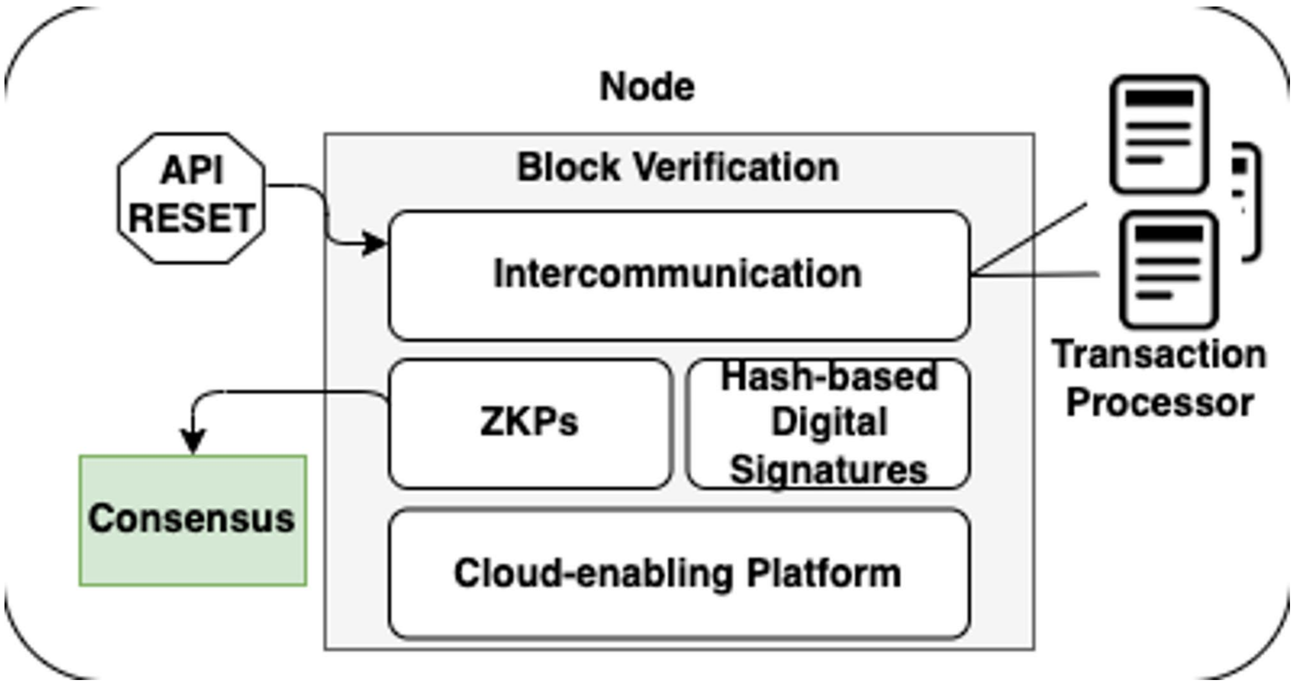

Fig. 5 BDLT Operations and their Execution

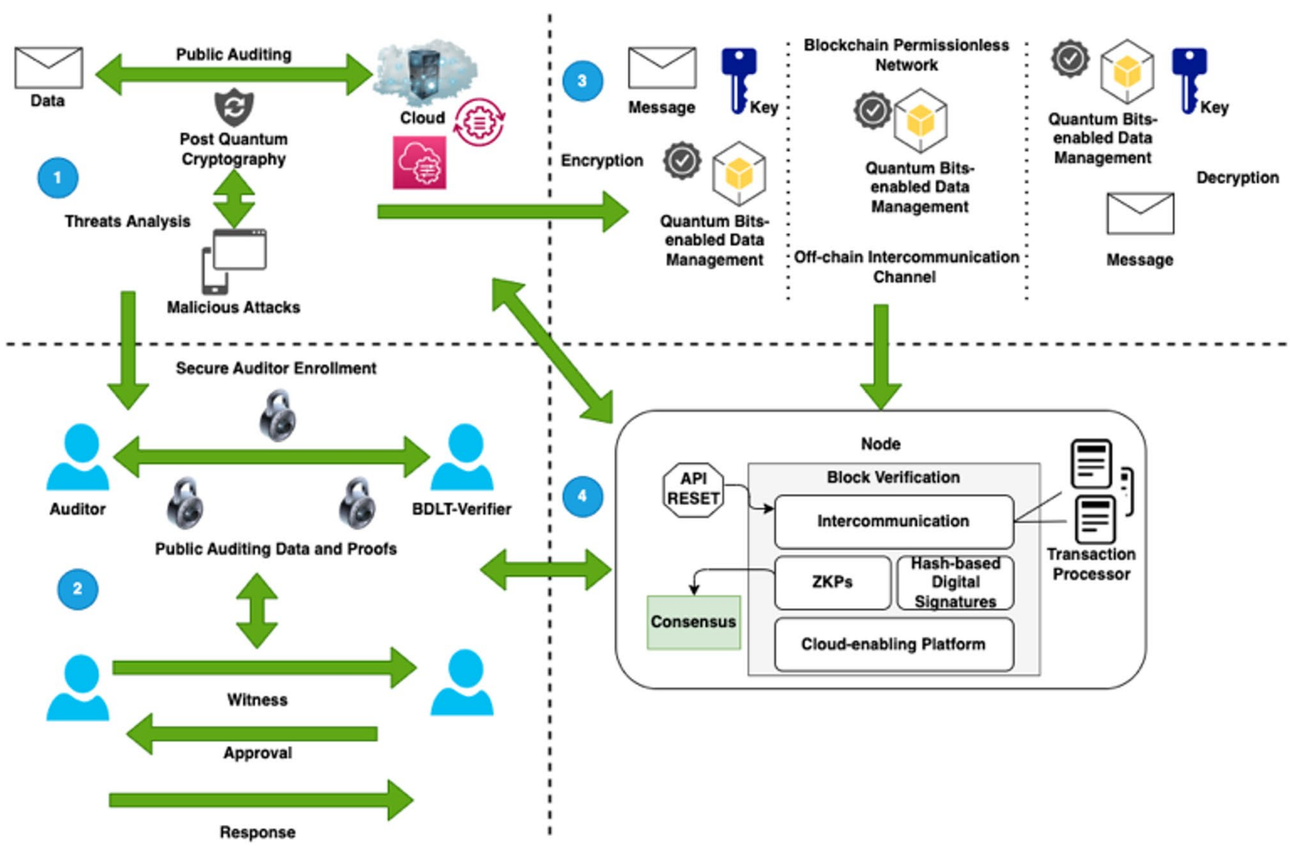

Fig. 6 The Proposed Framework

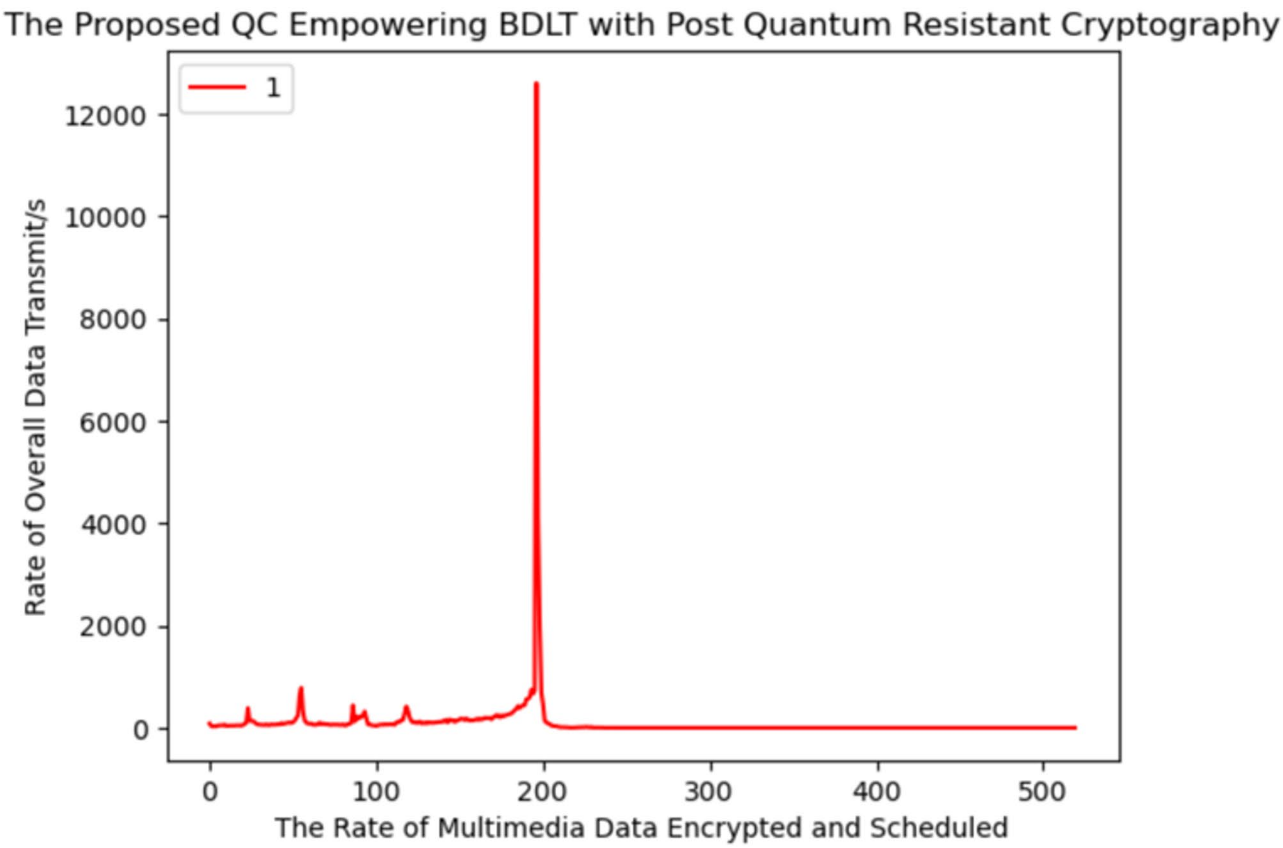

Fig. 7 Simulation Results (1), Multimedia Data Encryption

first assesses the request for data examination; if there is no threat, access is then granted. The following is an analysis of post-quantum cryptography in this process: as stated and indicated that the multimedia data Md (data request) is the first to be encrypted using symmetric key cryptography C. The following is the design of the equation: [symmetric key (Md)] = C [Encrypt]. A lattice-based post-quantum encryption technique, such as Dilithium, which is shown as follows, is then used to protect this symmetric key, Ck = Post Quantum Key Generation (symmetric key) * Post Quantum Encryption. While their hash digest is recorded on the BDLT based on the following criteria the integration of C and Ck is maintained in the cloud, H (C | Ck | T) = hash (Md).

Confidentiality and integrity are ensured while the second stage, the ZKPs, operates even in the scenario of quantum adversarial models. To let a third-party auditor to evaluate and verify the data without having access to Md, a non-interactive ZKP is made (data request). This is seen as a favourable result, and it provides the following lightweight and cost-effective environment: Proof Management (PM) = ZKPs (hash (Md), Sigma). Validation/Verification (hash (Md), PM) * ZKPs, as indicated in the third stage of the suggested solution, can be evaluated and confirmed by the public audit. This tuning and its flexibility provide confidentiality and integrity while upholding accountability.

Simulation, results, discussion

Prior to executing the suggested solution's simulation test, the system need must be adjusted in accordance with the following standards:

- **Qubits**: A large number of quantum information is needed for multimedia data processing.
- **Scalability**: The QPU is scalable to accommodate the large database of multimedia data processing.
- **Interconnectivity**: The qubits are interconnected in order to allow the efficient quantum operations.
- **Quantum Error Correction**: While operating, the system can manage noise and decoherence.
- **Multimedia Encryption**: Post-Quantum cryptography provides enhanced security for multimedia data while transmit.
- **Role of Participating Stakeholder and Policies**: Participating stakeholders, including auditor, BDLT verifier, expert of the system, and BDLT administrators, must have policies (read only and read/write) according to their roles designed in this process.

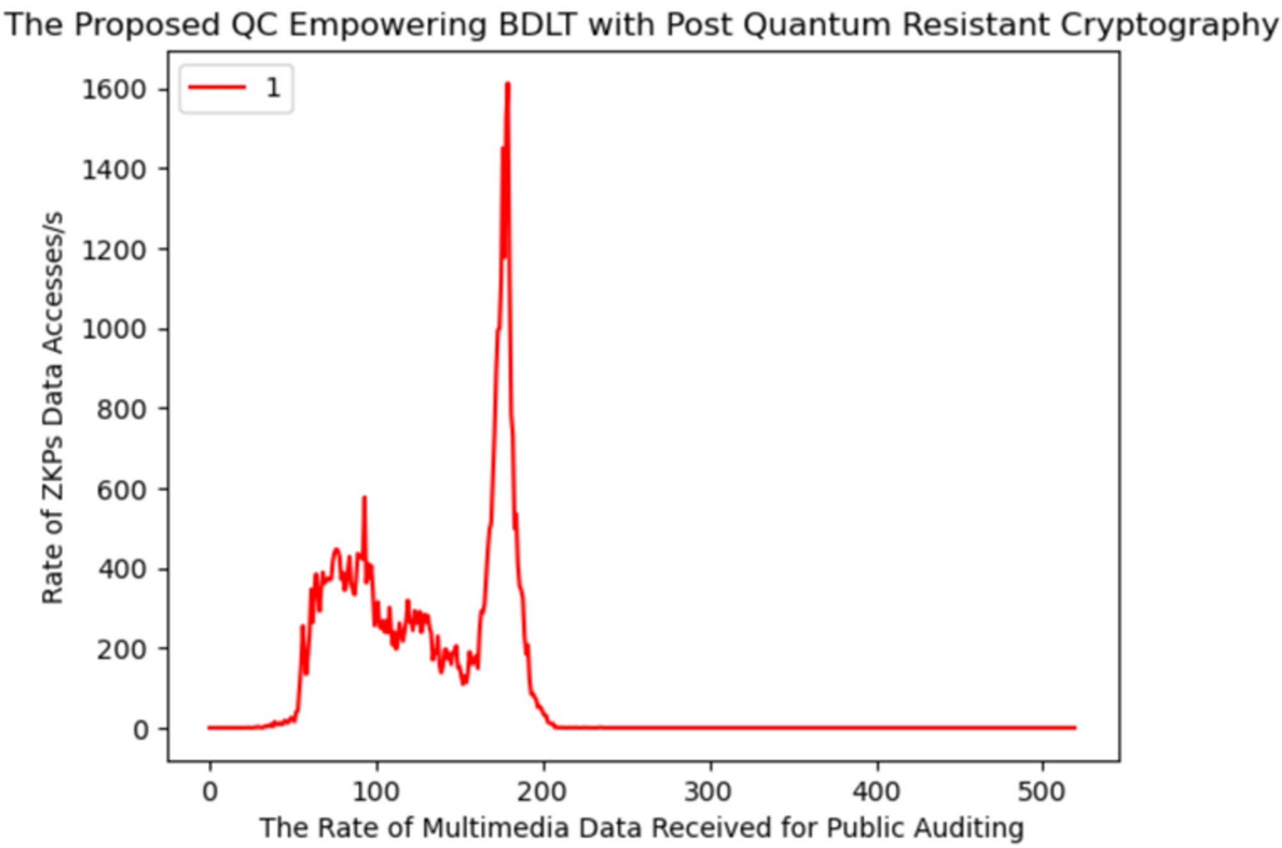

Fig. 8 Simulation Results (2), Multimedia Data Received for Public Auditing

- **Potential Attacks**: the list of potential attacks is mentioned as follows:
  - Harvest attacks.
  - Decrypted attacks.

The proposed QC enabling BDLT with post-quantum cryptography has successfully completed a simulation test, as shown in Fig. 7. The evaluation metrics are the rate of multimedia data encrypted and scheduled over the data transmission per second. But within 12,000 s, more than 500 of the data we get is encrypted. Conversely, using ZPKs data accessing/s, we are able to successfully test the simulation of the multimedia data received for public auditing. In contrast, 559 files in 1600 s of multimedia data were received for auditing, a shown in Fig. 8; all of this data was fully encrypted because ZKPs were included for affordable authentication.

But the purpose of the simulation test (3) is to compare the amount of safe and protected cloud preservation received every second with the amount of multimedia data audited and stored in cloud storage. The result shows that the proposed solution present a promising result in terms of 591 multimedia data files audited and preserved in the cloud storage units in less than of 2500, which is the total of 0.2364 files/s preserved and optimized, as shown in Fig. 9.

Along with that, Fig. 10 illustrates the designed to compare the amount of multimedia data audited and transmitted over the BDLT-enabled public network environment with the amount of secure and protected cloud preservation that was received every second. The outcome demonstrates that the proposed solution offers a promising outcome in terms of 544 multimedia data files audited and sent in the BDLT public network environment in less than 3500, or 0.1554 files/s optimized and communicated.

Similar to this, Fig. 11 shows how the proposed QC gives BDLT post-quantum cryptography in order to evaluate the rate of tamper-proof data preservation throughout the entire amount of data stored in the cloud, where B represents the blockchain integrated with “P” (post-quantum cryptography). We get excellent results, such as 273 untouched data files that were assessed and saved in the cloud storage in 12,000 s, or 0.022 data files stored per second.

The usefulness of the proposed framework is demonstrated by simulation results, which show that it reduces quantum vulnerability by 96.84%, improves auditing speed by 87.85%, and verifies data integrity with

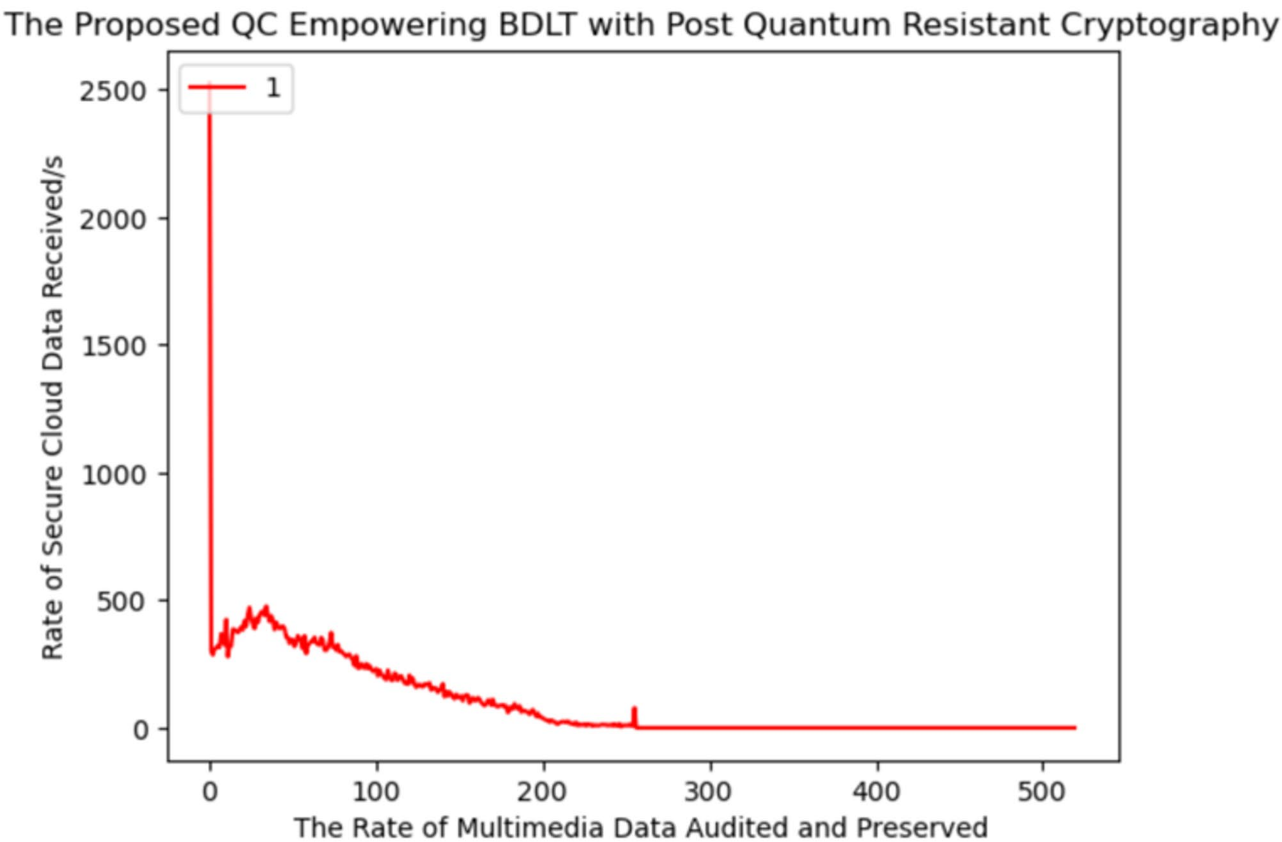

Fig. 9 Simulation Results (3), Multimedia Data Audited and Preserved in the Cloud

an accuracy of 98.21% when compared to traditional BDLT-enabled platforms. Furthermore, the framework's resilience is demonstrated by the 92.47% reduction in privacy leakage in multimedia systems, as shown in Fig. 12.

We conduct a comparative analysis of the proposed hybrid framework, which combines BDLT, post-quantum cryptography, and cloud computing for privacy-protected public auditing trials. The analysis is based on the solution for cloud data security and privacy, especially the major context is multimedia data privacy protection and preservation to make public auditing platforms more reliable and effective, as highlighted in Tables 2 and 3 (as illustrated in Fig. 13).

While traditional BDLT solutions are efficient and have massive adaptation, they still face severe risks from QC. Integrating post-quantum cryptography ensures future-proof security but introduces a performance trade-off that must be addressed to ensure scalability, as discussed in Tables 3 and 4, and Table 5.

Implementation challenges and limitations

Throughout this, the paper introduces a novel solution in terms of a quantum resilient, privacy preservation framework for cloud-based public auditing platforms; different implementation issues remain to address in future investigations. The list of implementation challenges and limitations are discussed as follows:

A. *Post-Quantum Cryptographic Scalability in Cloud Environment.*

There are several promising security features are available that are offered by post-quantum cryptographic mechanisms [34]; their performance on a large scale is better than the classical ones, especially in the cloud environment. At the same time, various hierarchies still remain an unresolved problem [35–38]. However, key size and encrypted latency in the lattice-based algorithm, such as Dilithium, can significantly hinder scalability in cloud platforms handling large amounts of multimedia content. Similarly, the existing BDLT-enabled consensus protocols, such as Proof of Authority (PoA), rely heavily on traditional cryptographic algorithms. As QC advances, these consensus protocols are likely to be vulnerable to quantum attacks, one of the major problems counted as a load of redundant object computation. Designing and implementing quantum-resilient BDLT-enabled consensus protocols that integrate lattice-based cryptographic primitives

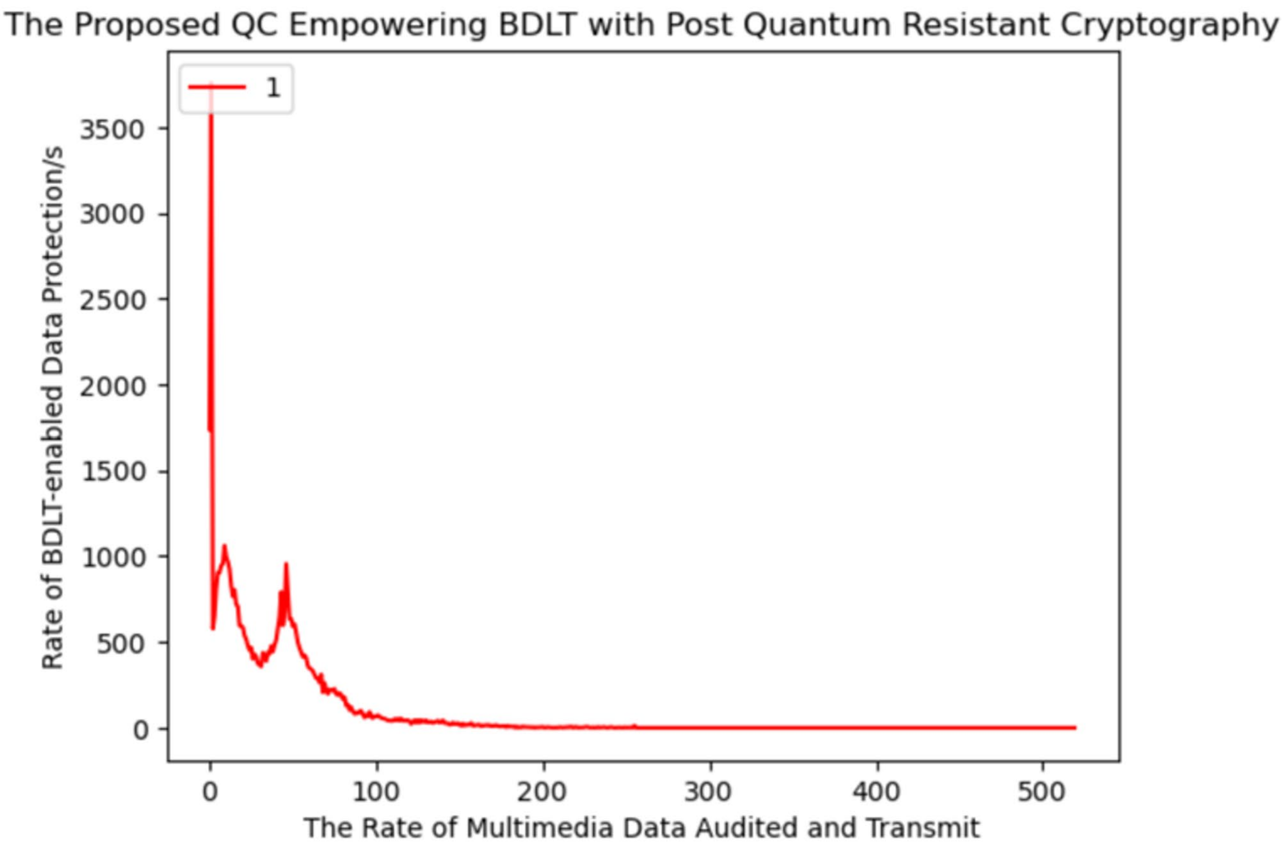

Fig. 10 Simulation Results (4), Multimedia Data Audited and Transmitted

with additional efficient features is still a critical open challenge.

The list of future research that derivates from this implementation issue is mentioned as follows:

- The management of local maxima and minima between key generation and organization, along with ciphertext storage for large datasets.
- Creating a lightweight post-quantum cryptographic method.
- Designing quantum secure consensus protocols.
- Futuristic investigation on hybrid approach of classical method with cryptographic post-resistant quantum technique.

B. Privacy Preservation of Auditing Techniques for Large-Scale Multimedia Data.

Due to their computational complexity in certifying big multimedia files, ZKPs and homomorphic encryption have demonstrated substantial value in privacy-preservation public auditing [37–39]. However, with the advent of QC and the cooperation of quantum machine learning models, public auditing platforms may benefit from an improved environment for malicious analysis. Effective hostile activity in BDLT and multimedia content is necessary for quantum-enhanced anomaly detection in order to enable classical algorithms.

The list of future research that derivates from this implementation issue is mentioned as follows [39–41]:

- Distributed auditing models need to be effectively designed.
- Efficient ZKP mechanism need to be implemented for large multimedia contents.
- More exploration requires on quantum machine learning for real-time anomaly detection purposes.

III.*Multi-Cloud, Edge Computing, and Fog-Based Cross Platform Implication.*

The proliferation of edge, fog, and multi-cloud infrastructures adds another level of complexity in ensuring quantum-resistant cryptographic security across distributed systems. Quantum safe cryptography is being used in fog node, edge node, and multi-cloud settings, such as for the transport and preservation of audiovisual material alone [42–47]. This issue remains a crucial one in this industry. On the other hand, developers must embrace new security standards and procedures as a result of the shift to quantum-safe cryptography

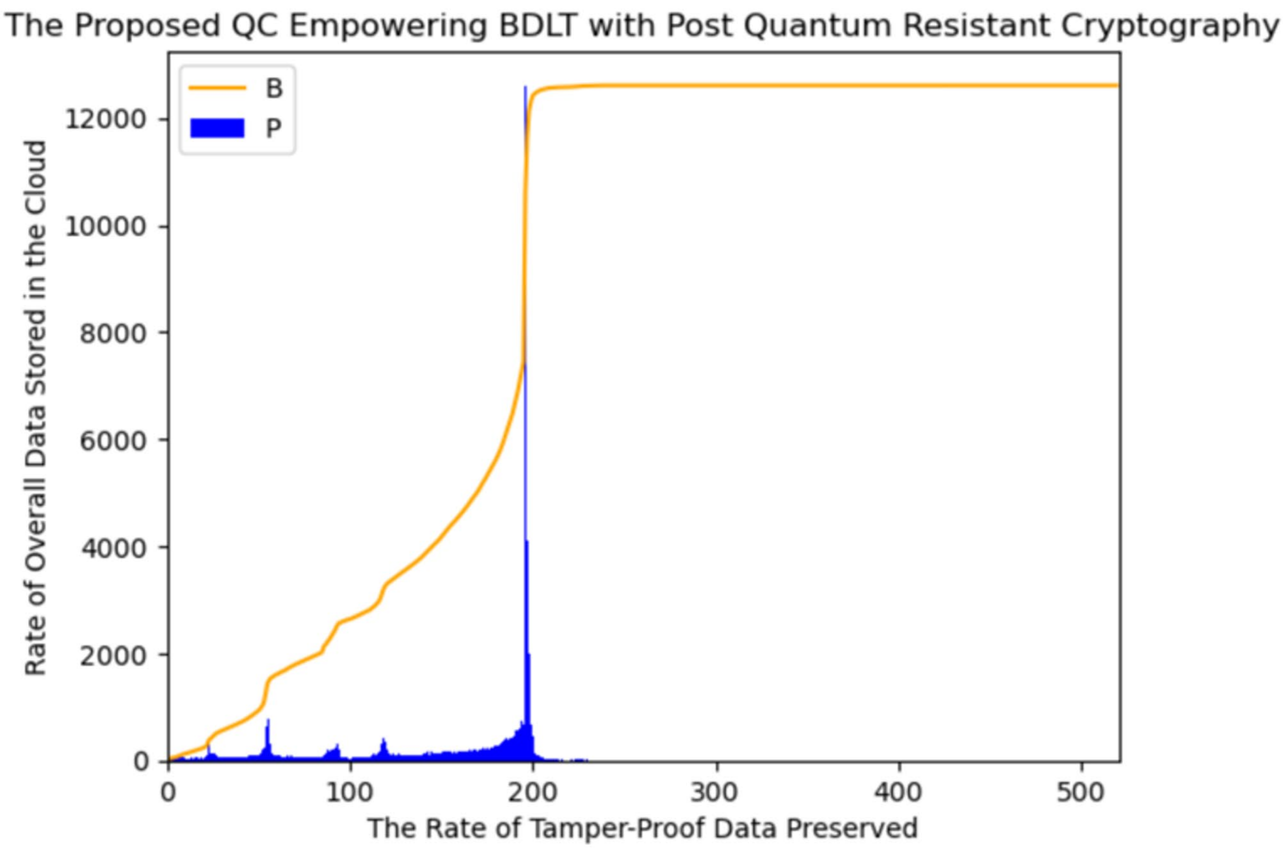

Fig. 11 Simulation Results (5), Rate of Tamper-Proof Data Preserved in the Cloud

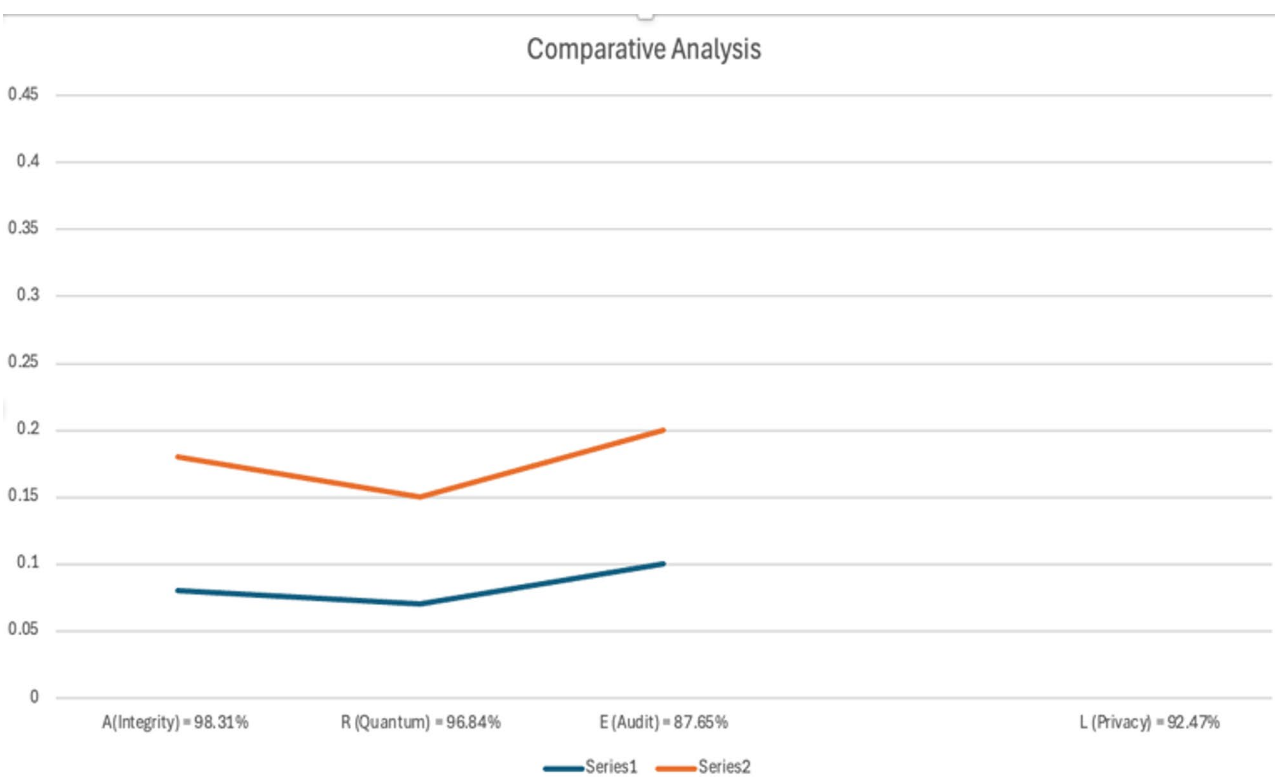

Fig. 12 Result of Comparative Analysis (1)

Table 2 Comparative analysis (1)

Comparative Constraints	BDLT with Traditional Cryptography	BDLT with Post-Quantum Cryptography
Quantum Attacks	√	√
Cryptographic Algorithms	√	√
Cost-Efficient Performance	X	√
Scalability	X	√

Table 3 Comparative analysis (2)

Comparative Constraints	Classical Public Auditing	BDLT-enabled Privacy Preservation and Auditing
Compliances	X	√
Audit Efficiency	√	√
Computational Overhead	√	√
Data Integrity Verification	X	√

Table 4 Comparative analysis (3)

Comparative Constraints	Blockchain-enabled Public Cryptography	Hybrid BDLT Framework with Post-Quantum Cryptography
Data Protection	√	√
Access Control	√	√
Cost of Integration	X	√
Use-Case Suitability	√	√

Table 5 Comparative analysis (4)

Comparative Constraints	Traditional Encryption Mechanism	Quantum Resilient Cryptography
Real-Time Auditing	X	√
Privacy	√	√
Preservation Protection	√	√
Scalability	X	√

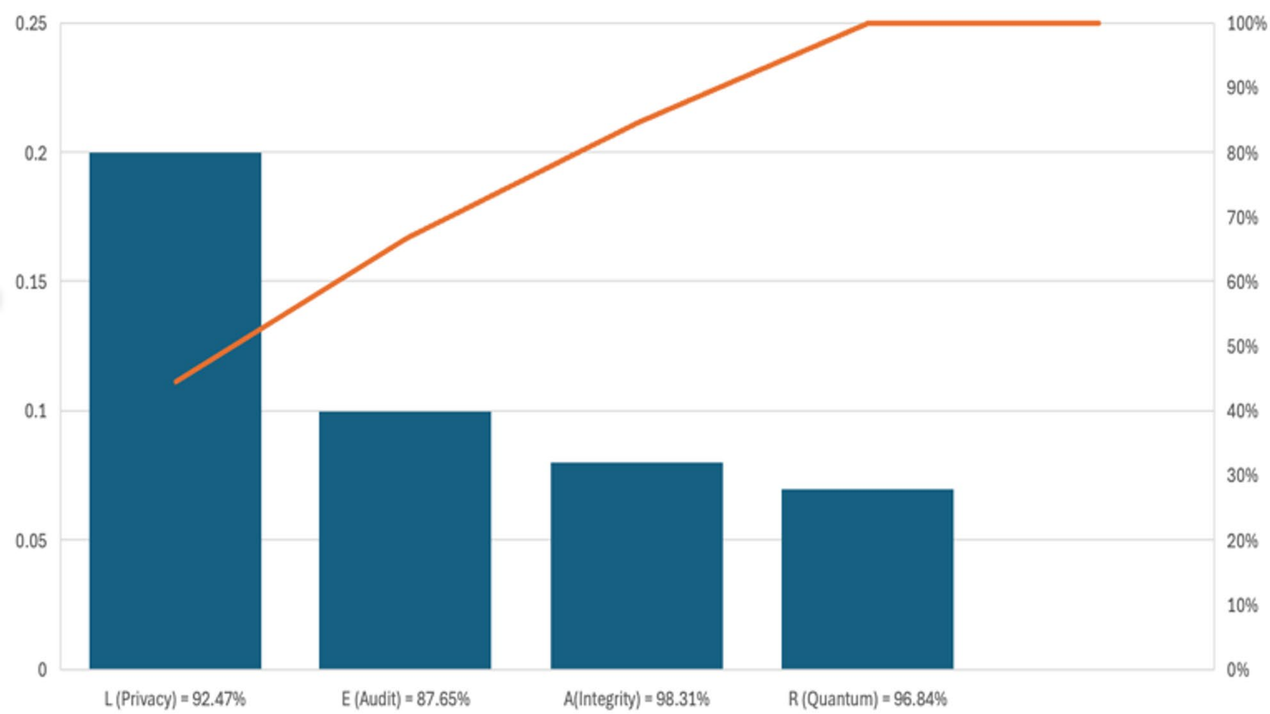

Fig. 13 Result of Comparative Analysis (2)

and related protocols. Use innovative quantum-resilient applications to provide a user-friendly environment and regulations.

The list of future research that derivates from this implementation issue is mentioned as follows:

- Cross platform-based cloud auditing protocols.
- Secure data exchange and preservation hierarchies.
- Investigation requires in user-enabled quantum secure cryptography acceptance.

Conclusion and future direction

This paper discussed the existing problems raised in the implication of post quantum resistance cryptographic algorithms. In order to achieve this, this research present a novel and secure framework that integrates BDLT, quantum resistant cryptography and privacy preservation in public auditing purposes. The major criteria are to secure multimedia data in cloud-enabling platforms. Although decentralized trust and transparency are provided by Blockchain Distributed Ledger Technology

(BDLT), the emergence of Quantum Computing poses a challenge to traditional cryptographic primitives, rendering multimedia data more susceptible. For the purpose of guaranteeing long-term data integrity and privacy preservation in cloud-based infrastructures, this study suggests an innovative and secure framework that combines BDLT with quantum-resilient, widely recognized post-quantum cryptographic algorithms. The benefit of the proposed solution is mentioned as follows:

- Provide a quantum resistant against future computing advancements.
- Provide scalability in BDLT and cryptographic algorithm for multimedia content.
- Provide an enhanced privacy through ZKPs while maintaining the efficiency of public auditing.

While the presented technique helps greatly to the development of quantum resistant BDLT Dapps for cloud data privacy. Nonetheless, the following research gaps remain unfilled:

- Optimization of post-quantum cryptographic algorithms.
- Scalability in BDLT with Hyperledger technology, for post-quantum computing environment.
- Integration of Quantum Machine Learning (QML) for threat evaluation and mitigation.
- Multi-cloud and edge computing integration for multimedia systems.
- Usability and compliance in Quantum Resilient Platforms.

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Author contributions
Abdullah Ayub Khan performs the Original Writing Part, Software, and Methodology; Abdullah Ayub Khan, Asif Ali Laghari, Hamad Almansour, Leila Jamel, Fahima Hajjej, Vania V. Estrela, Mohamad Afendee Mohamed, Sajid Ullah performs Rewriting, investigation, design Methodology, and Conceptualization; Abdullah Ayub Khan, Asif Ali Laghari, Hamad Almansour, Leila Jamel, Fahima Hajjej, Vania V. Estrela, Mohamad Afendee Mohamed, Sajid Ullah performs related work part and manage results and discussions; Abdullah Ayub Khan, Asif Ali Laghari, Hamad Almansour, Leila Jamel, Fahima Hajjej, Vania V. Estrela, Mohamad Afendee Mohamed, Sajid Ullah performs related work part and manage results and discussion; Abdullah Ayub Khan perform Rewriting, design Methodology, and Visualization; Abdullah Ayub Khan, Mohamad Afendee Mohamed performs Rewriting, design Methodology, and Visualization.

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Data availability
All data generated or analysed during this study are included in this published article.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
The committee of the Bahria University Karachi Campus confirmed that all experimental protocols were approved by the organization.

Ethical guidelines/accordance
It is confirmed that the experiments follow the criteria of ethics approval and consent to participates.

Informed consent
No human subjects are harmed in this research, and we confirmed that all data shared with the participants. The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Permission to reproduce material from other sources
Not applicable.

Conflict of interest
This paper's author addressed the fact that there is no conflict between them.

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